SUPPORTING ADULT GUIDANCE

Reasoning from Nets

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner moves fluently between a net and the solid it folds into: identifying the solid from its net, reasoning about opposite faces, testing which arrangements of squares close into a cube, and constructing nets of their own. The emphasis is on **the face-to-face mapping** between flat and solid — the 2D→3D bridge that makes surface area possible next lesson.

Driving Question: *How does a flat shape fold up into a solid — and which flat shapes will?*

The mathematics, in plain terms

A **net** is a flat arrangement of a solid's faces, joined along edges, that folds up into the solid. Folding only swings each face along an edge, so the faces never change size — which is why the area of a net equals the surface area of the solid. Each face of the net maps to exactly one face of the solid, and you can reason out which faces end up **opposite** one another (on a standard dice, opposite faces add to 7). A cube has six square faces, so its net is six squares joined edge to edge — but only certain arrangements close up: there are exactly **eleven** nets of a cube. A cuboid also has six faces (three matching pairs); a triangular prism has two triangular ends and three rectangles; a square-based pyramid has a square base and four triangles.

Each face has a destination

The most useful idea to protect is that folding a net is not guesswork: each flat face has a fixed destination on the solid. A learner who can point at a face and say where it will land — and which face it will sit opposite — is reasoning, not hoping. The commonest difficulty is deciding whether an arrangement will actually close; the repair is always to fold it, mentally or on screen, one face at a time and watch for two faces trying to land on the same side. Encourage the learner to check their own nets by folding rather than by memorising the eleven cube nets.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Predict & Fold: working out, from a net, which solid it makes and which face lands opposite a chosen one.	~15 min
Movement	The Net Folder: dragging a flat net up into a cube, cuboid, triangular prism or square-based pyramid, and rotating it.	~8 min
Reflection	'Will it close?' — deciding which arrangements of six squares fold into a cube, and why.	~10 min

Creativity	Design-a-Net: building your own cube net and watching it fold, aiming to find several of the eleven.	~5 min
Rest	A pause to notice which face was easiest, and which hardest, to place when folding.	~4 min
Nutrition	Contextual (Mode A): cardboard packaging — a cereal box is the net of a cuboid; a Toblerone box the net of a triangular prism.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking any six squares make a cube

A 2-by-3 block of squares does not fold into a cube. Repair: “Fold it in your mind one square at a time — do two faces try to land on the same side?”

Losing track of hidden faces

Forgetting the faces that end up at the back or underneath. Repair: “When it is folded, which faces can you not see from the front? Have you accounted for all six?”

Muddling opposite and adjacent faces

Repair: “Opposite faces never share an edge. On the net, which face never touches this one?”

Thinking folding changes the area

Repair: “When a face swings up, does it get any bigger or smaller? So how does the net's area compare with the solid's surface?”

Assuming every solid has one net

Repair: “How many different nets of a cube can you find? Is there really only one right answer?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the faces of a solid, naming 2D shapes, and the area of a rectangle. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the reasoning — and the folding — with the learner:

- “If you fold this up, which face ends up opposite the shaded one?”
- “Will this arrangement close into a cube? Fold it in your mind, one face at a time.”
- “Which faces can you not see when it is folded? Have you counted all six?”
- “How many different nets of this solid can you design?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to memorise the eleven cube nets?”

No. Far better that they can *test* any arrangement by folding it, mentally or on screen. Recognising why one closes and another does not is the real skill.

“Why does a net matter for later work?”

A net lays every face out flat, so it is exactly how surface area is found next lesson — add up the areas of all the faces in the net.

“They can picture a cube but not a pyramid or prism.”

Use the Net Folder and the isometric grid on the Scratchpad, and handle a real box. Building and unfolding a physical package is the strongest way to make it concrete.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.